SPECULA • DIGITAL FORENSIC EXAMINATION

Technical Incident Examination Report

CYBER DEFENSE / INCIDENT RESPONSE

Case LOCAL-c653ec6838174caabe0845b9a723dda8

1. Mandate, scope and document control

Field	Recorded value
Case identifier	LOCAL-c653ec6838174caabe0845b9a723dda8
Declared matter/title	Not recorded
Declared examiner	Not recorded
Declared organization	Not recorded
Declared examination authority	Not recorded
Investigation start (UTC)	2026-10-13T06:26:30.179267+00:00
Document generated (UTC)	2026-10-13T06:27:04.074148+00:00
Evidence completeness	incomplete
Normalized snapshot SHA-256	e9ad227be974d9c2dc2f8e7db9b84dd51e48fd4d9b684196c5c2503a75d08588

Scope is limited to the preserved records linked to this case. Supplied metadata is declarative. No conclusion about activity outside this collection window or unsupported log types is made.

2. Executive examination result

20 normalized evidence records and 14 evidence-cited detection findings were available. An absence of supported detections does not establish that a device is uncompromised.

Measure	Value
Case state	guardrail

Measure	Value
Acceptance status	Not recorded
Earliest recorded event	2025-01-01T00:00:00+00:00
Latest recorded event	2026-10-13T03:22:00+00:00
Executed agent steps	16

3. Evidence receipt, integrity and custody

Item	Recorded value
Collection source	uploaded_zip
Original archive SHA-256	d679313df39c4b4e6fdccef7685ed08fe259f84b8c2fcecce67b46af21320734
Archive byte length	5378
Receipt time (UTC)	2026-10-13T06:26:21.431391+00:00
Custody scope	Receipt recorded; acquisition and prior custody are uploader declarations

Original file	Bytes	SHA-256	Record count
edr/events.jsonl	909	594a2ae67003791958f420c05b382496d4824ef545d532334841dd9dd5fbb96c	3
gcp_audit/events.json	801	f3eae391e5fd30721920d6edd93996645b76f86f6938f61887e0c58bd7174477	2
malware/findings.json	535	99cf588b8e9d3966d9e20e58affadf6fee12d9545469b1cf92c961619f30e950	1
memory/findings.json	519	3d00a8ef5af1b292ae7cc34b4b92a0384c37ce3f78ade5d0249d644563e0a632	2
mft/records.json	614	a7155b6ecf08305f33c9d0399519c6a5efa7446ff0275025af1dc52b7e0fc899	1
ueba/events.json	751	aaca257c798307cfeebb973bc028ad67c283eabfc9e1a9f971ea204220d9099a	3

Original file	Bytes	SHA-256	Record count
vuln_scan/findings.json	475	1f6c1c49b639933580d534135a0c00f1e8f383a685e8d62f069d758db08ddac4	2
windows/Security.json	1606	7df5eaf5952ec760fa81a35eb4f2a94a2dc814db25227155dfb0ae5196e8642c	5
windows/Sysmon.jsonl	1696	17ea47ae3af61e0c1b2823d6acd490526484919eb156f8cfd7ca3a5c62560ef1	5

Original archive/file hashes identify bytes at receipt. Normalized record hashes identify derived data. They do not prove authenticity at acquisition, independent custody verification or legal admissibility. Hash verification on every subsequent transfer remains a reviewer responsibility.

Uploader-declared acquisition and custody


```
{
  "case_id": "SPECULA-TEST-001",
  "created_at": "2026-10-13T04:46:12.595124+00:00",
  "description": "Synthetic DFIR fixture for Specula pipeline validation. Contains intentional attack
indicators and anti-forensic tampering.",
  "examiner": "OpenAI synthetic fixture generator",
  "expected_attack_story": {
    "expected_tampering": [
      "Windows Security log clearing via wevtutil",
      "MFT timestamp inconsistency/timestomping",
      "malware executable staged under ProgramData"
    ],
    "expected_techniques": [
      "T1110",
      "T1078",
      "T1059.001",
      "T1003",
      "T1071.001",
      "T1041",
      "T1070.001",
      "T1070.006"
    ],
    "summary": "External credential attack -> successful remote logon -> PowerShell staging -> malware
drop/execution -> LSASS access -> C2/exfiltration -> log clearing -> timestomping; separate suspicious GCP
service-account activity."
  },
  "files": {
    "edr/events.jsonl": {
      "sha256": "594a2ae67003791958f420c05b382496d4824ef545d532334841dd9dd5fbb96c",
      "size_bytes": 909
    },
    "gcp_audit/events.json": {
      "sha256": "f3eae391e5fd30721920d6edd93996645b76f86f6938f61887e0c58bd7174477",
      "size_bytes": 801
    },
```

```
    "malware/findings.json": {
      "sha256": "99cf588b8e9d3966d9e20e58affadf6fee12d9545469b1cf92c961619f30e950",
      "size_bytes": 535
    },
    "memory/findings.json": {
      "sha256": "3d00a8ef5af1b292ae7cc34b4b92a0384c37ce3f78ade5d0249d644563e0a632",
      "size_bytes": 519
    },
    "mft/records.json": {
      "sha256": "a7155b6ecf08305f33c9d0399519c6a5efa7446ff0275025af1dc52b7e0fc899",
      "size_bytes": 614
    },
    "ueba/events.json": {
      "sha256": "aaca257c798307cfeebb973bc028ad67c283eabfc9e1a9f971ea204220d9099a",
      "size_bytes": 751
    },
    "vuln_scan/findings.json": {
      "sha256": "1f6c1c49b639933580d534135a0c00f1e8f383a685e8d62f069d758db08ddac4",
      "size_bytes": 475
    },
    "windows/Security.json": {
      "sha256": "7df5eaf5952ec760fa81a35eb4f2a94a2dc814db25227155dfb0ae5196e8642c",
      "size_bytes": 1606
    },
    "windows/Sysmon.jsonl": {
      "sha256": "17ea47ae3af61e0c1b2823d6acd490526484919eb156f8cfd7ca3a5c62560ef1",
      "size_bytes": 1696
    }
  },
  "synthetic": true
}
```


4. Methods and reproducibility

1. Preserve submitted bytes and inventory original files.
2. Apply security screening and supported source normalization to OCSF records with deterministic UIDs.
3. Read case-linked evidence; apply deterministic triage and supported detection rules.
4. Record model-assisted synthesis and debate as analysis requiring verification.
5. Render factual exhibits from saved structured records and retain incompleteness flags.

Runtime	Version
Python	3.10.22
Report template	specula-examination-v1

Source files measured at investigation start


```
{
  "packages": {
    "fastapi": "0.142.2",
```

```
    "langgraph": "1.2.12",
    "neo4j": "6.3.1",
    "pydantic": "2.13.5"
  },
  "python": "3.10.22",
  "source_sha256": {
    "src/agents/examination_reports.py": "155a7ffde81e3051fc126b305460032e99c636737316285b7bdeafff75295d1d",
    "src/agents/grounded_report.py": "ac80d7f6bd2e7d595fe1727694123a50ff484bb1fcca7e3dcc46b98c53b2af68",
    "src/agents/log_analysis/detection_rules.py":
"75e9c30a1064af27323b5c47be3d6bb4849b982c47321b90b572aa625828ee2f",
    "src/agents/log_analysis_factory.py":
"8823373498dc583aac9e0b9120ff2b0645bd0c1c85d5e4da6ff86ab3cff0a7e2",
    "src/ingestion/run_pipeline.py": "212afcdde783e71a90a1f43fef438b0d0b9b385c740b7e0588971fb344168077"
  }
}
```


Current report-rendering environment


```
{
  "packages": {
    "fastapi": "0.142.2",
    "langgraph": "1.2.12",
    "neo4j": "6.3.1",
    "pydantic": "2.13.5"
  },
  "python": "3.10.22",
  "source_sha256": {
    "src/agents/examination_reports.py": "155a7ffde81e3051fc126b305460032e99c636737316285b7bdeafff75295d1d",
    "src/agents/grounded_report.py": "ac80d7f6bd2e7d595fe1727694123a50ff484bb1fcca7e3dcc46b98c53b2af68",
    "src/agents/log_analysis/detection_rules.py":
"75e9c30a1064af27323b5c47be3d6bb4849b982c47321b90b572aa625828ee2f",
    "src/agents/log_analysis_factory.py":
"8823373498dc583aac9e0b9120ff2b0645bd0c1c85d5e4da6ff86ab3cff0a7e2",
    "src/ingestion/run_pipeline.py": "212afcdde783e71a90a1f43fef438b0d0b9b385c740b7e0588971fb344168077"
  }
}
```


Current source fingerprints do not prove which source version executed a historical investigation.

Timestamp conversion does not independently verify source clocks. Local evidence hashing and storage do not establish Hyperledger Fabric anchoring. Model outputs are not admitted as measured evidence.

5. Recorded automated detection results

Automated screening results require examiner validation. Citations establish record membership, not independent correctness of an inference.

log_analysis

Deterministic unusual_parent_process signal: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe; powershell.exe -NoProfile -ExecutionPolicy Bypass -EncodedCommand UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=; user=administrator

Evidence references: b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9

log_analysis

Command attempts to clear the Windows Security log; successful deletion is not established.

Evidence references: e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d

log_analysis

Executable launched from ProgramData: C:\ProgramData\svchost-helper.exe. Location alone does not establish malware.

Evidence references: ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130

log_analysis

PowerShell executed an encoded command; content and intent require review.

Evidence references: b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9

log_analysis

File creation timestamps disagree: SI=2025-01-01T00:00:00Z, FN=2026-10-13T03:10:00Z. Possible timestomping; copying/restoration must be ruled out.

Evidence references: 4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99

cloud_container

GCP service-account key creation observed for svc-backup@specula-lab-synthetic.iam.gserviceaccount.com; authorization and intent require review.

Evidence references: 4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c

identity

Source reports access to lsass.exe; completed credential extraction is not established.

Evidence references: 633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c

identity

2 failed logons followed by a successful logon for administrator from 203.0.113.45. Possible credential compromise; brute force is not established by this count alone.

Evidence references: 8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c, efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3, 05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3

insider_threat

Source-reported behavioral anomaly: remote_login; insider identity or intent is not established.

Evidence references: 781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e

insider_threat

Source-reported behavioral anomaly: large_outbound_transfer; insider identity or intent is not established.

Evidence references: abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2

malware_stylometry

Source reports an unsigned executable: Unsigned executable launched from ProgramData. No malware-family attribution is established.

Evidence references: 05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7

malware_stylometry

Source reports an unsigned executable: Unsigned executable launched from ProgramData. No malware-family attribution is established.

Evidence references: 2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d

memory_forensics

Source-reported memory finding: Credential material pattern observed in memory snapshot

Evidence references: 351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b

memory_forensics

Source-reported memory finding: RWX private memory region with PE-like header

Evidence references: 7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83

6. Event chronology and exhibit index

Exhibit hashes below identify normalized JSON, not original source-file bytes.

Exhibit / UID	Time (UTC)	OCSF class	Recorded observation	Normalized JSON SHA-256
E-00001 / 4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99	2025-01-01T00:00:00+00:00	1001	Normalized event	8669238ef61d8c30af235cfbba68e6f37c831664e16eb3b6a4d1b07a6fc8f9b1
E-00002 / 054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144	2026-10-13T03:00:00+00:00	3002	Normalized event	88daa1366d651463628badb0928ef50075f6b37a175a55edbd1041aa25726553
E-00003 / f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077	2026-10-13T03:00:00+00:00	2004	Normalized event	de1847772e388674e5fe1606b40670a3590b39d5485f59c32c6e891194614a26
E-00004 / 8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c	2026-10-13T03:05:00+00:00	3002	Normalized event	b0952b72b1082f0609cc255079670e3c0aea4bbabbf5d979d277661d1d68b757
E-00005 / efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3	2026-10-13T03:06:00+00:00	3002	Normalized event	3445dbecdbb00ba88d0ee39fa2b181846d548ba535a40c04f6f586f6f4c9bb87
E-00006 / 05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3	2026-10-13T03:07:00+00:00	3002	Normalized event	8032d18e3a9a0661705d17debfd183f98dcc7984ebc449b2632736789cf5655d
E-00007 / 781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e	2026-10-13T03:07:00+00:00	2004	Normalized event	912c1a2d5dd81ab610b95e184d631a2fa0984eea5fb08727dde3ac02b97b532b

Exhibit / UID	Time (UTC)	OCSF class	Recorded observation	Normalized JSON SHA-256
E-00008 / b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9	2026-10-13T03:08:00+00:00	1007	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	c303af80a642f2514879f29f4f886f4cab2670b9e9c09e448244b95c9920f5ae
E-00009 / fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d	2026-10-13T03:10:00+00:00	1001	Normalized event	c6a5e510e26d8540ba46869d8ae46e7be1b26372d4a9cab267f37bdcce53acbe
E-00010 / 5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50	2026-10-13T03:11:00+00:00	3002	Normalized event	63cbd66d9361d8afd677bfe91b491ed1373e6caa9cdacc2539be6c957f345c49
E-00011 / 05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7	2026-10-13T03:12:00+00:00	1007	svchost-helper.exe	177624dd86ee15186dbb7504ef66dc80d208f082192a64cfdf8c2057b5aab8a3
E-00012 / 2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d	2026-10-13T03:12:00+00:00	2004	Normalized event	1cd74b6a4eee8c9213b0b8da26d6e43087e767901b037557273b3eec44cdc020
E-00013 / ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130	2026-10-13T03:12:00+00:00	1007	C:\ProgramData\svchost-helper.exe	e2fa1c02acc6030a1569ff1851f4433cdacead776f80afa660a99dd78bbf2547
E-00014 / 351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b	2026-10-13T03:13:00+00:00	2004	Normalized event	c91773dfd5b992edad279165183b82e5e9fb75ea836bb9a67df8d87a2d3cae6b
E-00015 / 633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c	2026-10-13T03:13:00+00:00	2004	Normalized event	14bcc8a7b33ce573ab9bf68cb809940c25625a85961c29721ab713edd42a5fd0

Exhibit / UID	Time (UTC)	OCSF class	Recorded observation	Normalized JSON SHA-256
E-00016 / 7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83	2026-10-13T03:13:00+00:00	2004	Normalized event	02601789229b3084c9a2a91028d006f4a7f99a2679802b735433dfae233cdc34
E-00017 / abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2	2026-10-13T03:14:00+00:00	2004	Normalized event	8ae5e6e6dcc3c81ee64834859dd90924fa49109b7277485f4789e530b6c97d49
E-00018 / e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d	2026-10-13T03:15:00+00:00	1007	C:\Windows\System32\cmd.exe	3495634ddd43dcc88f487b3649073c5e1a956ad1a4a8da915a6d59c3e8cd8cc5
E-00019 / 4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c	2026-10-13T03:20:00+00:00	6003	Normalized event	6ec1b5ac9124fddcb789b9ff3aab19bc681a05021fcbe42db5ef202072a82f63
E-00020 / 060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c	2026-10-13T03:22:00+00:00	6003	Normalized event	246b030c32e8011b3d0a92c0f599383e8031cc5bd64b6817b4927ad70915f08d

7. Telemetry coverage and ingestion diagnostics


```
{
  "class_counts": {
    "1001": 2,
    "1007": 4,
    "2004": 7,
    "3002": 5,
    "6003": 2
  },
  "guardrails": {
    "guardrail_tier1_result": "pass",
    "guardrail_tier2_result": "pass",
    "guardrail_tier3_result": "pass",
    "hitl_decision": null
  },
  "processing": {
    "coverage": "Only supported valid records are normalized; unsupported and filtered records may be
excluded",
    "normalization_contract": "uploaded-export-aliases-v1",
```

```
    "normalized_events": 20,
    "record_errors_sample": [
      {
        "file": "edr/events.jsonl",
        "reason": "Unsupported event type, security rejection, validation failure or excluded port",
        "record": 3
      },
      {
        "error_type": "ValueError",
        "file": "malware/findings.json",
        "reason": "Missing event timestamp; supply timestamp (ISO 8601 with UTC offset)",
        "record": 1
      },
      {
        "error_type": "ValueError",
        "file": "vuln_scan/findings.json",
        "reason": "Missing event timestamp; supply timestamp (ISO 8601 with UTC offset)",
        "record": 1
      },
      {
        "error_type": "ValueError",
        "file": "vuln_scan/findings.json",
        "reason": "Missing event timestamp; supply timestamp (ISO 8601 with UTC offset)",
        "record": 2
      },
      {
        "file": "windows/Sysmon.jsonl",
        "reason": "Unsupported event type, security rejection, validation failure or excluded port",
        "record": 2
      }
    ],
    "selected_records": 24,
    "selected_records_without_normalized_events": 5,
    "timestamp_policy": "Source times only; MFT modified-time alias uses standard_information_modified when
no native event time exists"
  },
  "selection": {
    "end_date": "",
    "excluded_ports": [
      80,
      443,
      53
    ],
    "max_events_per_file": 2000,
    "start_date": ""
  },
  "triage": [
    {
      "reason_code": "default_keep",
      "uid": "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
      "verdict": "KEEP"
    },
    {
      "reason_code": "high_severity",
      "uid": "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
      "verdict": "ESCALATE"
    },
    {
      "reason_code": "default_keep",
```

```
    "uid": "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
    "verdict": "KEEP"
  },
  {
    "reason_code": "default_keep",
    "uid": "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
    "verdict": "KEEP"
  },
  {
    "reason_code": "high_severity",
    "uid": "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
    "verdict": "ESCALATE"
  },
  {
    "reason_code": "high_severity",
    "uid": "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
    "verdict": "ESCALATE"
  },
  {
    "reason_code": "default_keep",
    "uid": "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
    "verdict": "KEEP"
  },
  {
    "reason_code": "default_keep",
    "uid": "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
    "verdict": "KEEP"
  },
  {
    "reason_code": "default_keep",
    "uid": "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
    "verdict": "KEEP"
  },
  {
    "reason_code": "high_severity",
    "uid": "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
    "verdict": "ESCALATE"
  },
  {
    "reason_code": "high_severity",
    "uid": "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
    "verdict": "ESCALATE"
  },
  {
    "reason_code": "default_keep",
    "uid": "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
    "verdict": "KEEP"
  },
  {
    "reason_code": "default_keep",
    "uid": "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
    "verdict": "KEEP"
  },
  {
    "reason_code": "default_keep",
    "uid": "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
    "verdict": "KEEP"
  },
  {
```

```
      "reason_code": "default_keep",
      "uid": "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
      "verdict": "KEEP"
    },
    {
      "reason_code": "default_keep",
      "uid": "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
      "verdict": "KEEP"
    },
    {
      "reason_code": "default_keep",
      "uid": "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
      "verdict": "KEEP"
    },
    {
      "reason_code": "default_keep",
      "uid": "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
      "verdict": "KEEP"
    },
    {
      "reason_code": "default_keep",
      "uid": "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
      "verdict": "KEEP"
    },
    {
      "reason_code": "default_keep",
      "uid": "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d",
      "verdict": "KEEP"
    }
  ]
}
```


8. Observed activity graph


```
{
  "edges": [
    {
      "dfkg_refs": [
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      ],
      "id": "edge-4",
      "source": "observation-2",
      "target": "observation-4",
      "type": "PRECEDES"
    },
    {
      "dfkg_refs": [
        "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      ],
      "id": "edge-5",
      "source": "observation-4",
      "target": "observation-5",
      "type": "PRECEDES"
    },
```

```
    {
      "dfkg_refs": [
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130"
      ],
      "id": "edge-8",
      "source": "observation-7",
      "target": "observation-8",
      "type": "PRECEDES"
    },
    {
      "dfkg_refs": [
        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
      ],
      "id": "edge-11",
      "source": "observation-10",
      "target": "observation-11",
      "type": "PRECEDES"
    },
    {
      "dfkg_refs": [
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
      ],
      "id": "edge-12",
      "source": "observation-11",
      "target": "observation-12",
      "type": "PRECEDES"
    }
  ],
  "exploit_path_status": "unavailable",
  "kind": "observed_activity",
  "limitations": [
    "PRECEDES is temporal association, not causation.",
    "No validated exploit topology or calibrated probabilities available; no risk probability computed."
  ],
  "nodes": [
    {
      "dfkg_refs": [
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99"
      ],
      "id": "observation-0",
      "label": "timestamp_inconsistency",
      "summary": "File creation timestamps disagree: SI=2025-01-01T00:00:00Z, FN=2026-10-13T03:10:00Z.
Possible timestomping; copying/restoration must be ruled out.",
      "time": "2025-01-01T00:00:00Z"
    },
    {
      "dfkg_refs": [
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3"
      ],
      "id": "observation-1",
      "label": "failed_logons_followed_by_success",
      "summary": "2 failed logons followed by a successful logon for administrator from 203.0.113.45.
Possible credential compromise; brute force is not established by this count alone.",
      "time": "2026-10-13T03:07:00Z"
```

```
    },
    {
      "dfkg_refs": [
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e"
      ],
      "id": "observation-2",
      "label": "source_behavior_anomaly",
      "summary": "Source-reported behavioral anomaly: remote_login; insider identity or intent is not
established.",
      "time": "2026-10-13T03:07:00Z"
    },
    {
      "dfkg_refs": [
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      ],
      "id": "observation-3",
      "label": "unusual_parent_process",
      "summary": "Deterministic unusual_parent_process signal:
C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe; powershell.exe -NoProfile -ExecutionPolicy
Bypass -EncodedCommand UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=; user=administrator",
      "time": "2026-10-13T03:08:00+00:00"
    },
    {
      "dfkg_refs": [
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      ],
      "id": "observation-4",
      "label": "encoded_powershell",
      "summary": "PowerShell executed an encoded command; content and intent require review.",
      "time": "2026-10-13T03:08:00Z"
    },
    {
      "dfkg_refs": [
        "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7"
      ],
      "id": "observation-5",
      "label": "source_unsigned_executable",
      "summary": "Source reports an unsigned executable: Unsigned executable launched from ProgramData. No
malware-family attribution is established.",
      "time": "2026-10-13T03:12:00Z"
    },
    {
      "dfkg_refs": [
        "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d"
      ],
      "id": "observation-6",
      "label": "source_unsigned_executable",
      "summary": "Source reports an unsigned executable: Unsigned executable launched from ProgramData. No
malware-family attribution is established.",
      "time": "2026-10-13T03:12:00Z"
    },
    {
      "dfkg_refs": [
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130"
      ],
      "id": "observation-7",
      "label": "unusual_executable_location",
      "summary": "Executable launched from ProgramData: C:\\ProgramData\\svchost-helper.exe. Location alone
does not establish malware.",
```

```
      "time": "2026-10-13T03:12:00Z"
    },
    {
      "dfkg_refs": [
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c"
      ],
      "id": "observation-8",
      "label": "source_credential_access",
      "summary": "Source reports access to lsass.exe; completed credential extraction is not established.",
      "time": "2026-10-13T03:13:00Z"
    },
    {
      "dfkg_refs": [
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b"
      ],
      "id": "observation-9",
      "label": "source_memory_finding",
      "summary": "Source-reported memory finding: Credential material pattern observed in memory snapshot",
      "time": "2026-10-13T03:13:00Z"
    },
    {
      "dfkg_refs": [
        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83"
      ],
      "id": "observation-10",
      "label": "source_memory_finding",
      "summary": "Source-reported memory finding: RWX private memory region with PE-like header",
      "time": "2026-10-13T03:13:00Z"
    },
    {
      "dfkg_refs": [
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
      ],
      "id": "observation-11",
      "label": "source_behavior_anomaly",
      "summary": "Source-reported behavioral anomaly: large_outbound_transfer; insider identity or intent is
not established.",
      "time": "2026-10-13T03:14:00Z"
    },
    {
      "dfkg_refs": [
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
      ],
      "id": "observation-12",
      "label": "security_log_clear_attempt",
      "summary": "Command attempts to clear the Windows Security log; successful deletion is not
established.",
      "time": "2026-10-13T03:15:00Z"
    },
    {
      "dfkg_refs": [
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c"
      ],
      "id": "observation-13",
      "label": "service_account_key_creation",
      "summary": "GCP service-account key creation observed for svc-backup@specula-lab-
synthetic.iam.gserviceaccount.com; authorization and intent require review.",
      "time": "2026-10-13T03:20:00Z"
    }
```

```
  ]
}
```


9. Agent execution, synthesis and hypotheses

Model-generated hypotheses and attack-path prose are unverified unless supported by independently checked evidence. No structured exploit path is certified by prose.


```
{
  "agent_traces": [
    {
      "action": "case_evidence_review",
      "agent_role": "supervisor",
      "dfkg_refs": [
        "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
        "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
        "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
        "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
        "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
        "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
        "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
      ],
      "latency_ms": 8127.4,
      "model_output": "{\"selected_ids\":[1,4,5,7,9,10,11,12,13,14,15,16,17,18,19]}",
      "model_output_validated": true,
      "model_used": "qwen3:8b",
      "observation": "Reviewed 20 cited observations. Model selection validated.",
      "selected_observation_ids": [
        1,
        4,
        5,
        7,
        9,
        10,
        11,
        12,
        13,
        14,
        15,
        16,
        17,
        18,
```

```
        19
      ],
      "timestamp": "2026-10-13T06:26:38.782090+00:00"
    },
    {
      "action": "case_scoped_retrieval_and_triage",
      "agent_role": "evidence_collection",
      "latency_ms": 15.8,
      "model_used": "deterministic_rules",
      "observation": "Case evidence collection complete: 20/20 fetched events triaged; 5 escalated, 15
kept.",
      "terminal": true,
      "termination_reason": "complete",
      "thought": "Triage actual case events"
    },
    {
      "action": "run_on_batch",
      "agent_role": "log_analysis",
      "latency_ms": 1511.5,
      "model_invocations": 1,
      "model_output": "The event batch [1] indicates an unusual parent process, as signaled by the
'unusual_parent_process' pattern. No prior findings are available for citation.",
      "model_output_validated": false,
      "model_used": "qwen3:8b",
      "observation": "Produced 5 findings.",
      "thought": "Analyzed 6 log events."
    },
    {
      "action": "deterministic_network_analysis",
      "agent_role": "network_forensics",
      "latency_ms": 0,
      "model_used": "deterministic_rules",
      "observation": "Produced 0 evidence-backed findings; 0 partial batches.",
      "thought": "Read 0 OCSF 4001 events across 0 hosts."
    },
    {
      "action": "case_evidence_review",
      "agent_role": "cloud_container",
      "dfkg_refs": [
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c"
      ],
      "latency_ms": 1955.5,
      "model_output": "{\"selected_ids\":[0]}",
      "model_output_validated": true,
      "model_used": "qwen3:8b",
      "observation": "Reviewed 1 cited observations. Model selection validated.",
      "selected_observation_ids": [
        0
      ],
      "timestamp": "2026-10-13T06:26:42.547753+00:00"
    },
    {
      "action": "case_evidence_review",
      "agent_role": "identity",
      "dfkg_refs": [
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3"
```

```
    ],
    "latency_ms": 2764.0,
    "model_output": "{\"selected_ids\":[0,1]}",
    "model_output_validated": true,
    "model_used": "qwen3:8b",
    "observation": "Reviewed 2 cited observations. Model selection validated.",
    "selected_observation_ids": [
      0,
      1
    ],
    "timestamp": "2026-10-13T06:26:43.358807+00:00"
  },
  {
    "action": "case_evidence_review",
    "agent_role": "insider_threat",
    "dfkg_refs": [
      "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
      "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
    ],
    "latency_ms": 1107.4,
    "model_output": "{\"selected_ids\":[0,1]}",
    "model_output_validated": true,
    "model_used": "qwen3:8b",
    "observation": "Reviewed 2 cited observations. Model selection validated.",
    "selected_observation_ids": [
      0,
      1
    ],
    "timestamp": "2026-10-13T06:26:41.704201+00:00"
  },
  {
    "action": "case_evidence_review",
    "agent_role": "malware_stylometry",
    "dfkg_refs": [
      "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
      "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d"
    ],
    "latency_ms": 2317.9,
    "model_output": "{\"selected_ids\":[0,1]}",
    "model_output_validated": true,
    "model_used": "qwen3:8b",
    "observation": "Reviewed 2 cited observations. Model selection validated.",
    "selected_observation_ids": [
      0,
      1
    ],
    "timestamp": "2026-10-13T06:26:42.916932+00:00"
  },
  {
    "action": "case_evidence_review",
    "agent_role": "memory_forensics",
    "dfkg_refs": [
      "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
      "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83"
    ],
    "latency_ms": 1640.1,
    "model_output": "{\"selected_ids\":[0,1]}",
    "model_output_validated": true,
    "model_used": "qwen3:8b",
```

```
      "observation": "Reviewed 2 cited observations. Model selection validated.",
      "selected_observation_ids": [
        0,
        1
      ],
      "timestamp": "2026-10-13T06:26:42.242655+00:00"
    },
    {
      "action": "case_evidence_review",
      "agent_role": "timeline_reconstruction",
      "dfkg_refs": [
        "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
        "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
        "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
        "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
        "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
        "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
        "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
      ],
      "latency_ms": 2827.7,
      "model_output": "{\"selected_ids\":[0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19]}",
      "model_output_validated": true,
      "model_used": "qwen3:8b",
      "observation": "Reviewed 20 cited observations. Model selection validated.",
      "selected_observation_ids": [
        0,
        1,
        2,
        3,
        4,
        5,
        6,
        7,
        8,
        9,
        10,
        11,
        12,
        13,
        14,
        15,
        16,
        17,
        18,
        19
      ],
```

```
      "timestamp": "2026-10-13T06:26:46.208925+00:00"
    },
    {
      "action": "deterministic_attack_profile_scoring",
      "agent_role": "threat_attribution",
      "algorithm_version": "jaccard-sw-v2",
      "candidate_scores": [
        {
          "actor_id": "G0039",
          "combined_score": 0.11428571428571428
        },
        {
          "actor_id": "G0011",
          "combined_score": 0.08000000000000002
        },
        {
          "actor_id": "G0033",
          "combined_score": 0.08000000000000002
        }
      ],
      "corpus_hash": "14caeb8656d3b8379bb89c0ba1e6b540d25ebdce2cb687a97cb2d010a8c320a0",
      "corpus_version": "ATT&CK-xlsx-v19.2",
      "degraded_flags": [
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      "T1059.001:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
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        "technique_id": "T1003",
        "threat_intel_refs": [
          "T1003:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
        ],
        "time": "2026-10-13T03:13:00Z"
      }
    ],
    "overall_confidence": 0.0222,
    "scoring_parameters": {
      "gap": -1,
      "inferred_sequence_weight": 0.0,
      "jaccard_weight": 0.4,
      "match": 2,
      "mismatch": -1,
      "smith_waterman_weight": 0.6
    },
    "summary": "Top ATT&CK profile is Suckfly (G0039; similarity 0.114). Evidence-adjusted confidence is
0.022 (bounds 0.000-0.272). This profile comparison does not establish actor identity. Profile sequence is
inferred from ATT&CK tactic order, not a recorded actor chronology. No actor sequence is available for this
comparison. The ranking compares the cited case TTPs with ATT&CK group profiles.",
    "threat_intel_refs": [
      "G0011:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "G0033:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "G0039:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "T1003:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "T1059.001:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "T1070.006:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "T1078:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
```

```
    ],
    "top_candidate": {
      "actor_id": "G0039",
      "combined_score": 0.11428571428571428,
      "jaccard": 0.2857142857142857,
      "matched_techniques": [
        "T1003",
        "T1078"
      ],
      "name": "Suckfly",
      "profile_techniques": [
        "T1003",
        "T1046",
        "T1059.003",
        "T1078",
        "T1553.002"
      ],
      "sequence_available": false,
      "sequence_basis": "attack_tactic_order",
      "smith_waterman": 0.0,
      "source_hash": "85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
      "source_version": "ATT&CK-xlsx-v19.2"
    }
  },
  "findings": [
    {
      "agent_role": "supervisor",
      "dfkg_refs": [],
      "kafka_offset": null,
      "summary": "Primary agents dispatched for 20 case events; specialist routing follows actual evidence
after collection.",
      "timestamp": "2026-10-13T06:26:38.782090+00:00"
    },
    {
      "agent_role": "evidence_collection",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
        "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
        "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
        "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
        "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
        "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
        "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
      ],
      "evidence_collection": {
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  "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
  "complete": true,
  "degraded": false,
  "degraded_flags": [],
  "dfkg_refs": [
    "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
    "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
    "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
    "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
    "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
    "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
    "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
    "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
    "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
    "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
    "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
    "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
    "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
    "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
    "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
    "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
    "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
    "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
    "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
    "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
  ],
  "fetched_count": 20,
  "loop_count": 0,
  "records": [
    {
      "activity_id": 1,
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
      "category_uid": 3,
      "class_uid": 3002,
      "clock_skew_offset_ms": 0,
      "clock_skew_unverified": false,
      "dst_endpoint": {
        "canonical_host_uid": "host-WS-17",
        "ip_address": null,
        "port": null
      },
      "is_summary": false,
      "logon_type": 2,
      "ocsf_version": "1.2.0",
      "raw_source_timestamp": "2026-10-13T03:00:00Z",
      "security_scan_degraded": false,
      "severity_id": 1,
      "source_event_id": 4624,
      "source_outcome": "success",
      "source_type": "evtx",
      "source_user_name": "alice",
      "src_endpoint": {
        "canonical_host_uid": null,
        "ip_address": "10.10.20.44",
        "port": null
      },
      "time": "2026-10-13T03:00:00Z",
      "trace_id": "f2eef222-b6b4-4c0b-bfba-16c643f9c920",
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```
    "uid": "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
    "user": {
      "domain": null,
      "name": "alice",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-e7cc792584f84b4792553c73cd53c73d",
    "category_uid": 1,
    "class_uid": 1007,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "file_path": "C:\\ProgramData\\svchost-helper.exe",
    "host_name": "WS-17",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "parent_process_pid": 0,
    "process_name": "svchost-helper.exe",
    "process_pid": 0,
    "raw_source_timestamp": "2026-10-13T03:12:00Z",
    "security_scan_degraded": false,
    "severity_id": 4,
    "source_event_type": "process_start",
    "source_finding": "Unsigned executable launched from ProgramData",
    "source_type": "edr",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:12:00Z",
    "trace_id": "cd659720-9d61-4f9f-a9f8-d5c49fb0ce2e",
    "uid": "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
    "user_name": "administrator"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "logon_type": 10,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:07:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4624,
    "source_outcome": "success",
    "source_type": "evtx",
    "source_user_name": "administrator",
    "src_endpoint": {
      "canonical_host_uid": null,
```

```
      "ip_address": "203.0.113.45",
      "port": null
    },
    "time": "2026-10-13T03:07:00Z",
    "trace_id": "9874d634-cb5f-475b-8789-8ac52a210917",
    "uid": "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "api_operation": "storage.objects.list",
    "canonical_host_id": "gcp-project-specula-lab-synthetic",
    "case_id": "LOCAL-e7cc792584f84b4792553c73cd53c73d",
    "category_uid": 6,
    "class_uid": 6003,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "cloud_account_id": "specula-lab-synthetic",
    "cloud_provider": "GCP",
    "cloud_region": "global",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:22:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_ip": "203.0.113.45",
    "source_type": "gcp_audit",
    "time": "2026-10-13T03:22:00Z",
    "trace_id": "ec0d789c-3829-4b74-8867-63621ddda158",
    "uid": "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
    "user_identity": "svc-backup@specula-lab-synthetic.iam.gserviceaccount.com"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "High",
    "finding_title": "Unsigned executable launched from ProgramData",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:12:00Z",
    "security_scan_degraded": false,
    "severity_id": 4,
    "source_event_type": "process_start",
    "source_finding": "Unsigned executable launched from ProgramData",
    "source_type": "edr",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:12:00Z",
    "trace_id": "60288c97-9df8-4e49-a331-b619035f7ec0",
    "uid": "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d"
  },
```

```
  {
    "activity_id": 1,
    "analytic_name": "Volatility windows.lsadump",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "LastRecordChange": "2026-10-13T03:13:00Z",
      "TimeCreated": "2026-10-13T03:13:00Z",
      "eventTime": "2026-10-13T03:13:00Z",
      "finding": "Credential material pattern observed in memory snapshot",
      "host": "WS-17",
      "host_name": "WS-17",
      "note": "Synthetic finding only; no real credentials present",
      "plugin": "windows.lsadump",
      "severity": "critical",
      "severity_id": 5,
      "sha256_hash": "996da47417d742ba5b8aa65c177d0f02ce0d496f3aedea3beca1b441756f4ff3",
      "timestamp": "2026-10-13T03:13:00Z",
      "trace_id": "06b3adcb-4e6a-4448-8adb-a2f82284fff3",
      "vct_merkle_root": "5d21ffa6395177977223d673b13865dabd01282ea85572e34728a9160459a0c7"
    },
    "finding_title": "Source-reported memory finding: windows.lsadump",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:13:00Z",
    "raw_verdict": "Credential material pattern observed in memory snapshot",
    "security_scan_degraded": false,
    "severity_id": 5,
    "source_finding": "Credential material pattern observed in memory snapshot",
    "source_type": "memory",
    "time": "2026-10-13T03:13:00Z",
    "trace_id": "06b3adcb-4e6a-4448-8adb-a2f82284fff3",
    "uid": "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b"
  },
  {
    "activity_id": 1,
    "api_operation": "google.iam.admin.v1.CreateServiceAccountKey",
    "canonical_host_id": "gcp-project-specula-lab-synthetic",
    "case_id": "LOCAL-e7cc792584f84b4792553c73cd53c73d",
    "category_uid": 6,
    "class_uid": 6003,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "cloud_account_id": "specula-lab-synthetic",
    "cloud_provider": "GCP",
    "cloud_region": "global",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:20:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_ip": "203.0.113.45",
    "source_type": "gcp_audit",
    "time": "2026-10-13T03:20:00Z",
```

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    "trace_id": "c11a8b82-0ee1-47b4-a02f-6e4822d5f0d8",
    "uid": "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
    "user_identity": "svc-backup@specula-lab-synthetic.iam.gserviceaccount.com"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-LOCAL_HOST",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1001,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "file_name": "svchost-helper.exe",
    "file_path": "C:\\ProgramData",
    "fn_created": "2026-10-13T03:10:00Z",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2025-01-01T00:00:00+00:00",
    "security_scan_degraded": true,
    "severity_id": 1,
    "si_created": "2025-01-01T00:00:00Z",
    "source_type": "mft",
    "source_volume_creation_time": "2026-08-01T00:00:00+00:00",
    "time": "2025-01-01T00:00:00Z",
    "trace_id": "ce97e6eb-ffad-4bcc-ad9b-f5ada5e0509c",
    "uid": "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:11:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4672,
    "source_type": "evtx",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:11:00Z",
    "trace_id": "23188401-e454-4e98-b12d-5ae9f290ab52",
    "uid": "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
    "activity_id": 1,
```

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      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "category_uid": 2,
      "class_uid": 2004,
      "clock_skew_offset_ms": 0,
      "clock_skew_unverified": true,
      "confidence": "Unspecified",
      "details": {
        "access_mask": "0x1010",
        "attack_indicator": "credential dumping behavior",
        "event_type": "credential_access",
        "host": "WS-17",
        "operation": "OpenProcess",
        "process": "svchost-helper.exe",
        "severity": "critical",
        "sha256_hash": "3657b111b92214eaf6b42430d11fa883c3b410d792aa7a634c37fc39103d103a",
        "target_process": "lsass.exe",
        "timestamp": "2026-10-13T03:13:00Z",
        "trace_id": "aeff10aa-0b18-4fc3-9dd4-05b0b9b30103",
        "vct_merkle_root": "e3fc26a4b936c28eb5ab765e8d469dad3e97db5f1226e37d4e7cb8d81a089b7e"
      },
      "finding_title": "Source-reported credential access",
      "is_summary": false,
      "ocsf_version": "1.2.0",
      "raw_source_timestamp": "2026-10-13T03:13:00Z",
      "security_scan_degraded": false,
      "severity_id": 5,
      "source_event_type": "credential_access",
      "source_type": "edr",
      "time": "2026-10-13T03:13:00Z",
      "trace_id": "aeff10aa-0b18-4fc3-9dd4-05b0b9b30103",
      "uid": "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c"
    },
    {
      "activity_id": 1,
      "analytic_name": "Volatility windows.malfind",
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "category_uid": 2,
      "class_uid": 2004,
      "clock_skew_offset_ms": 0,
      "clock_skew_unverified": true,
      "confidence": "Unspecified",
      "details": {
        "ImageFileName": "svchost-helper.exe",
        "LastRecordChange": "2026-10-13T03:13:00Z",
        "PID": 4312,
        "TimeCreated": "2026-10-13T03:13:00Z",
        "eventTime": "2026-10-13T03:13:00Z",
        "finding": "RWX private memory region with PE-like header",
        "host": "WS-17",
        "host_name": "WS-17",
        "pid": 4312,
        "plugin": "windows.malfind",
        "process": "svchost-helper.exe",
        "process_name": "svchost-helper.exe",
        "process_pid": 4312,
        "severity": "high",
        "severity_id": 4,
```

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      "sha256_hash": "247f3c3322f5d3f164ac52f50773f841077ef579d10dccde7314d747ff928b96",
      "timestamp": "2026-10-13T03:13:00Z",
      "trace_id": "bd1687f5-3a34-4811-9897-762450cfa2b5",
      "vct_merkle_root": "d7cb0fafef5c4ae2c42fca3ca2c176ad4a4eba969f416615405d71dfc76dfc3e"
    },
    "finding_title": "Source-reported memory finding: windows.malfind",
    "is_summary": false,
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    "clock_skew_unverified": true,
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    "details": {
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      "artifact_type": "ueba_observation",
      "baseline_status": "anomalous",
      "confidence": "Unspecified",
      "event": "remote_login",
      "eventTime": "2026-10-13T03:07:00Z",
      "host": "WS-17",
      "host_name": "WS-17",
      "reasons": [
        "never-before-seen external source",
        "login after repeated failures"
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      "risk_score": 92,
      "severity_id": 1,
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      "source_ip": "203.0.113.45",
      "src_ip": "203.0.113.45",
      "timestamp": "2026-10-13T03:07:00Z",
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      "user": "administrator",
      "user_name": "administrator",
      "vct_merkle_root": "4c2b6d153342866f4c1ef1c2fcd17cae3a726513ef9f4ea6883f6522d8deca6a",
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    },
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    "source_event_type": "remote_login",
    "source_type": "ueba",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:07:00Z",
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    "uid": "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e"
  },
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    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
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      "port": null
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    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:05:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4625,
    "source_outcome": "failure",
    "source_type": "evtx",
    "source_user_name": "administrator",
    "src_endpoint": {
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      "ip_address": "203.0.113.45",
      "port": null
    },
    "time": "2026-10-13T03:05:00Z",
    "trace_id": "473f6c04-e4a0-43d8-899b-b746e794ed47",
    "uid": "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
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    "analytic_name": "UEBA Engine",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
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      "confidence": "Unspecified",
      "destination_ip": "198.51.100.77",
      "dst_ip": "198.51.100.77",
      "event": "large_outbound_transfer",
      "eventTime": "2026-10-13T03:14:00Z",
      "host": "WS-17",
      "host_name": "WS-17",
      "risk_score": 88,
      "severity_id": 1,
      "sha256_hash": "36c535ee3c88475f6d0a563266c91f4da94f95c69409b9f00ca67681d4fe3f93",
      "timestamp": "2026-10-13T03:14:00Z",
      "trace_id": "6ffd9f32-3ac0-47ff-bcdb-8eda321a6bce",
      "user": "administrator",
      "user_name": "administrator",
      "vct_merkle_root": "606e6ad44c98b9e3da78ecaab467c9b7d7b4e21427124f403b6c32357cfe431e",
      "verdict": "anomalous"
    },
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    "is_summary": false,
    "ocsf_version": "1.2.0",
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    "severity_id": 1,
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    "source_event_type": "large_outbound_transfer",
    "source_type": "ueba",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:14:00Z",
    "trace_id": "6ffd9f32-3ac0-47ff-bcdb-8eda321a6bce",
    "uid": "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1007,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "command_line": "C:\\ProgramData\\svchost-helper.exe --service",
    "host_name": "WS-17",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "parent_process_name": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
    "parent_process_pid": 0,
    "process_name": "C:\\ProgramData\\svchost-helper.exe",
    "process_pid": 0,
    "raw_source_timestamp": "2026-10-13T03:12:00Z",
    "security_scan_degraded": true,
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    "source_event_id": 1,
    "source_type": "evtx",
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          },
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            "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
            "category_uid": 1,
            "class_uid": 1007,
            "clock_skew_offset_ms": 0,
            "clock_skew_unverified": false,
            "command_line": "powershell.exe -NoProfile -ExecutionPolicy Bypass -EncodedCommand
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            "host_name": "WS-17",
            "is_summary": false,
            "ocsf_version": "1.2.0",
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            "parent_process_pid": 0,
            "process_name": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
            "process_pid": 0,
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            "security_scan_degraded": true,
            "severity_id": 1,
            "source_event_id": 1,
            "source_type": "evtx",
            "source_user_name": "administrator",
            "time": "2026-10-13T03:08:00Z",
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            "uid": "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
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            "class_uid": 1007,
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            "uid": "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
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    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
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    "source_event_id": 4625,
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    "source_type": "evtx",
    "source_user_name": "administrator",
    "src_endpoint": {
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      "ip_address": "203.0.113.45",
      "port": null
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      "sid": null
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    "class_uid": 2004,
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      "baseline_status": "normal",
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      "eventTime": "2026-10-13T03:00:00Z",
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      "host_name": "WS-17",
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        "verdict": "normal"
      },
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      "is_summary": false,
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      "raw_verdict": "normal",
      "security_scan_degraded": false,
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      "source_baseline_status": "normal",
      "source_event_type": "login",
      "source_type": "ueba",
      "source_user_name": "alice",
      "time": "2026-10-13T03:00:00Z",
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    },
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      "class_uid": 1001,
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      "clock_skew_unverified": false,
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      "is_summary": false,
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      "source_type": "evtx",
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    }
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    "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
    "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
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    "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
    "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
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      "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
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      "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
      "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
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        "verdict": "KEEP"
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        "uid": "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
        "verdict": "ESCALATE"
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        "uid": "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
        "verdict": "KEEP"
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        "uid": "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
        "verdict": "KEEP"
      },
      {
        "reason_code": "high_severity",
        "uid": "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
        "verdict": "ESCALATE"
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        "verdict": "ESCALATE"
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        "verdict": "KEEP"
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        "verdict": "KEEP"
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        "verdict": "ESCALATE"
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      {
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            "uid": "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
            "verdict": "KEEP"
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            "uid": "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
            "verdict": "KEEP"
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            "verdict": "KEEP"
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            "verdict": "KEEP"
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        "triaged_count": 20
      },
      "kafka_offset": null,
      "summary": "Case evidence collection complete: 20/20 fetched events triaged; 5 escalated, 15 kept.",
      "timestamp": "2026-10-13T06:26:39.060789+00:00"
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    {
      "affected_count": 1,
      "agent_role": "log_analysis",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
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      "confidence": 0.85,
      "dfkg_refs": [
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      ],
      "event_time": "2026-10-13T03:08:00+00:00",
      "finding_type": "unusual_parent_process",
      "kafka_offset": 0,
      "kafka_publication": {
        "status": "pending"
      },
      "model_narrative": "The event batch [1] indicates an unusual parent process, as signaled by the
'unusual_parent_process' pattern. No prior findings are available for citation.",
      "pattern": "unusual_parent_process",
      "severity": "high",
      "source": "validated_case_rule",
      "summary": "Deterministic unusual_parent_process signal:
C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe; powershell.exe -NoProfile -ExecutionPolicy
Bypass -EncodedCommand UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=; user=administrator",
      "timeline_range": [
        "2026-10-13 03:08:00+00:00",
        "2026-10-13 03:08:00+00:00"
      ],
      "timestamp": "2026-10-13T03:08:00+00:00"
    },
    {
      "agent_role": "log_analysis",
      "attacks": [
        "T1070.001"
      ],
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
      ],
      "event_time": "2026-10-13T03:15:00Z",
      "finding_type": "security_log_clear_attempt",
      "pattern": "security_log_clear_attempt",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Command attempts to clear the Windows Security log; successful deletion is not
established.",
      "timestamp": "2026-10-13T03:15:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
    {
      "agent_role": "log_analysis",
      "attacks": [],
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130"
      ],
      "event_time": "2026-10-13T03:12:00Z",
      "finding_type": "unusual_executable_location",
      "pattern": "unusual_executable_location",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Executable launched from ProgramData: C:\\ProgramData\\svchost-helper.exe. Location alone
does not establish malware.",
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      "timestamp": "2026-10-13T03:12:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
    {
      "agent_role": "log_analysis",
      "attacks": [
        "T1059.001"
      ],
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      ],
      "event_time": "2026-10-13T03:08:00Z",
      "finding_type": "encoded_powershell",
      "pattern": "encoded_powershell",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "PowerShell executed an encoded command; content and intent require review.",
      "timestamp": "2026-10-13T03:08:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
    {
      "agent_role": "log_analysis",
      "attacks": [
        "T1070.006"
      ],
      "canonical_host_id": "host-LOCAL_HOST",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99"
      ],
      "event_time": "2025-01-01T00:00:00Z",
      "finding_type": "timestamp_inconsistency",
      "pattern": "timestamp_inconsistency",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "File creation timestamps disagree: SI=2025-01-01T00:00:00Z, FN=2026-10-13T03:10:00Z.
Possible timestomping; copying/restoration must be ruled out.",
      "timestamp": "2025-01-01T00:00:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
    {
      "agent_role": "cloud_container",
      "attacks": [],
      "canonical_host_id": "gcp-project-specula-lab-synthetic",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c"
      ],
      "event_time": "2026-10-13T03:20:00Z",
      "finding_type": "service_account_key_creation",
      "pattern": "service_account_key_creation",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "GCP service-account key creation observed for svc-backup@specula-lab-
synthetic.iam.gserviceaccount.com; authorization and intent require review.",
      "timestamp": "2026-10-13T03:20:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
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    },
    {
      "agent_role": "identity",
      "attacks": [
        "T1003"
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      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c"
      ],
      "event_time": "2026-10-13T03:13:00Z",
      "finding_type": "source_credential_access",
      "pattern": "source_credential_access",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Source reports access to lsass.exe; completed credential extraction is not established.",
      "timestamp": "2026-10-13T03:13:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
    {
      "agent_role": "identity",
      "attacks": [
        "T1078"
      ],
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3"
      ],
      "event_time": "2026-10-13T03:07:00Z",
      "finding_type": "failed_logons_followed_by_success",
      "pattern": "failed_logons_followed_by_success",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "2 failed logons followed by a successful logon for administrator from 203.0.113.45.
Possible credential compromise; brute force is not established by this count alone.",
      "timestamp": "2026-10-13T03:07:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
    {
      "agent_role": "insider_threat",
      "attacks": [],
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e"
      ],
      "event_time": "2026-10-13T03:07:00Z",
      "finding_type": "source_behavior_anomaly",
      "pattern": "source_behavior_anomaly",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Source-reported behavioral anomaly: remote_login; insider identity or intent is not
established.",
      "timestamp": "2026-10-13T03:07:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
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    },
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      "agent_role": "insider_threat",
      "attacks": [],
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      "dfkg_refs": [
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
      ],
      "event_time": "2026-10-13T03:14:00Z",
      "finding_type": "source_behavior_anomaly",
      "pattern": "source_behavior_anomaly",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Source-reported behavioral anomaly: large_outbound_transfer; insider identity or intent is
not established.",
      "timestamp": "2026-10-13T03:14:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
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      "agent_role": "malware_stylometry",
      "attacks": [],
      "canonical_host_id": "host-WS-17",
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7"
      ],
      "event_time": "2026-10-13T03:12:00Z",
      "finding_type": "source_unsigned_executable",
      "pattern": "source_unsigned_executable",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Source reports an unsigned executable: Unsigned executable launched from ProgramData. No
malware-family attribution is established.",
      "timestamp": "2026-10-13T03:12:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
    },
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      "attacks": [],
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      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
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      ],
      "event_time": "2026-10-13T03:12:00Z",
      "finding_type": "source_unsigned_executable",
      "pattern": "source_unsigned_executable",
      "severity_id": 4,
      "source": "validated_case_rule",
      "summary": "Source reports an unsigned executable: Unsigned executable launched from ProgramData. No
malware-family attribution is established.",
      "timestamp": "2026-10-13T03:12:00Z",
      "trace_id": "8535320546564f68ba0d2f7f0b261590"
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      "agent_role": "memory_forensics",
      "attacks": [
        "T1003"
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    ],
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
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    ],
    "event_time": "2026-10-13T03:13:00Z",
    "finding_type": "source_memory_finding",
    "pattern": "source_memory_finding",
    "severity_id": 4,
    "source": "validated_case_rule",
    "summary": "Source-reported memory finding: Credential material pattern observed in memory snapshot",
    "timestamp": "2026-10-13T03:13:00Z",
    "trace_id": "8535320546564f68ba0d2f7f0b261590"
  },
  {
    "agent_role": "memory_forensics",
    "attacks": [],
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "dfkg_refs": [
      "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83"
    ],
    "event_time": "2026-10-13T03:13:00Z",
    "finding_type": "source_memory_finding",
    "pattern": "source_memory_finding",
    "severity_id": 4,
    "source": "validated_case_rule",
    "summary": "Source-reported memory finding: RWX private memory region with PE-like header",
    "timestamp": "2026-10-13T03:13:00Z",
    "trace_id": "8535320546564f68ba0d2f7f0b261590"
  },
  {
    "agent_role": "threat_attribution",
    "attribution_detail": {
      "algorithm_version": "jaccard-sw-v2",
      "candidate_actors": [
        {
          "actor_id": "G0039",
          "combined_score": 0.11428571428571428,
          "jaccard": 0.2857142857142857,
          "matched_techniques": [
            "T1003",
            "T1078"
          ],
          "name": "Suckfly",
          "profile_techniques": [
            "T1003",
            "T1046",
            "T1059.003",
            "T1078",
            "T1553.002"
          ],
          "sequence_available": false,
          "sequence_basis": "attack_tactic_order",
          "smith_waterman": 0.0,
          "source_hash": "85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "source_version": "ATT&CK-xlsx-v19.2"
        },
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      {
        "actor_id": "G0011",
        "combined_score": 0.08000000000000002,
        "jaccard": 0.2,
        "matched_techniques": [
          "T1078"
        ],
        "name": "PittyTiger",
        "profile_techniques": [
          "T1078",
          "T1588.002"
        ],
        "sequence_available": false,
        "sequence_basis": "attack_tactic_order",
        "smith_waterman": 0.0,
        "source_hash": "85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "source_version": "ATT&CK-xlsx-v19.2"
      },
      {
        "actor_id": "G0033",
        "combined_score": 0.08000000000000002,
        "jaccard": 0.2,
        "matched_techniques": [
          "T1003",
          "T1059.001"
        ],
        "name": "Poseidon Group",
        "profile_techniques": [
          "T1003",
          "T1007",
          "T1036.005",
          "T1049",
          "T1057",
          "T1059.001",
          "T1087.001",
          "T1087.002"
        ],
        "sequence_available": false,
        "sequence_basis": "attack_tactic_order",
        "smith_waterman": 0.0,
        "source_hash": "85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "source_version": "ATT&CK-xlsx-v19.2"
      }
    ],
    "confidence_basis": "deterministic_similarity_with_evidence",
    "confidence_lower": 0.0,
    "confidence_upper": 0.2722,
    "corpus_hash": "14caeb8656d3b8379bb89c0ba1e6b540d25ebdce2cb687a97cb2d010a8c320a0",
    "corpus_version": "ATT&CK-xlsx-v19.2",
    "degraded_flags": [
      "actor_sequence_inferred",
      "actor_sequence_unavailable",
      "semantic_mapping_unavailable",
      "unsupported_technique_id"
    ],
    "dfkg_refs": [
      "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
      "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
      "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
```

```
          "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
          "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
          "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
          "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3"
        ],
        "narrative": "Top ATT&CK profile is Suckfly (G0039; similarity 0.114). Evidence-adjusted confidence
is 0.022 (bounds 0.000-0.272). This profile comparison does not establish actor identity. Profile sequence
is inferred from ATT&CK tactic order, not a recorded actor chronology. No actor sequence is available for
this comparison. The ranking compares the cited case TTPs with ATT&CK group profiles.",
        "observed_ttps": [
          {
            "evidence_uids": [
              "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99"
            ],
            "technique_id": "T1070.006",
            "threat_intel_refs": [
              "T1070.006:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
            ],
            "time": "2025-01-01T00:00:00Z"
          },
          {
            "evidence_uids": [
              "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c"
            ],
            "technique_id": "T1078",
            "threat_intel_refs": [
              "T1078:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
            ],
            "time": "2026-10-13T03:05:00Z"
          },
          {
            "evidence_uids": [
              "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3"
            ],
            "technique_id": "T1078",
            "threat_intel_refs": [
              "T1078:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
            ],
            "time": "2026-10-13T03:06:00Z"
          },
          {
            "evidence_uids": [
              "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3"
            ],
            "technique_id": "T1078",
            "threat_intel_refs": [
              "T1078:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
            ],
            "time": "2026-10-13T03:07:00Z"
          },
          {
            "evidence_uids": [
              "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
            ],
            "technique_id": "T1059.001",
            "threat_intel_refs": [
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            ],
            "time": "2026-10-13T03:08:00Z"
```

```
          },
          {
            "evidence_uids": [
              "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b"
            ],
            "technique_id": "T1003",
            "threat_intel_refs": [
              "T1003:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
            ],
            "time": "2026-10-13T03:13:00Z"
          },
          {
            "evidence_uids": [
              "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c"
            ],
            "technique_id": "T1003",
            "threat_intel_refs": [
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            ],
            "time": "2026-10-13T03:13:00Z"
          }
        ],
        "overall_confidence": 0.0222,
        "scoring_parameters": {
          "gap": -1,
          "inferred_sequence_weight": 0.0,
          "jaccard_weight": 0.4,
          "match": 2,
          "mismatch": -1,
          "smith_waterman_weight": 0.6
        },
        "summary": "Top ATT&CK profile is Suckfly (G0039; similarity 0.114). Evidence-adjusted confidence is
0.022 (bounds 0.000-0.272). This profile comparison does not establish actor identity. Profile sequence is
inferred from ATT&CK tactic order, not a recorded actor chronology. No actor sequence is available for this
comparison. The ranking compares the cited case TTPs with ATT&CK group profiles.",
        "threat_intel_refs": [
          "G0011:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "G0033:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "G0039:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "T1003:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "T1059.001:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "T1070.006:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "T1078:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
        ],
        "top_candidate": {
          "actor_id": "G0039",
          "combined_score": 0.11428571428571428,
          "jaccard": 0.2857142857142857,
          "matched_techniques": [
            "T1003",
            "T1078"
          ],
          "name": "Suckfly",
          "profile_techniques": [
            "T1003",
            "T1046",
            "T1059.003",
            "T1078",
            "T1553.002"
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          ],
          "sequence_available": false,
          "sequence_basis": "attack_tactic_order",
          "smith_waterman": 0.0,
          "source_hash": "85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
          "source_version": "ATT&CK-xlsx-v19.2"
        }
      },
      "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
      "dfkg_refs": [
        "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
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      "kafka_publication": {
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        "status": "acknowledged",
        "topic": "findings.attribution"
      },
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      "threat_intel_refs": [
        "G0011:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "G0033:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "G0039:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "T1003:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "T1059.001:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "T1070.006:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02",
        "T1078:85a88d07176e67c504366c62e141b5be6e820581162ac5cddba28cf7b3d2bc02"
      ],
      "timestamp": "2026-10-13T06:26:49.254647+00:00",
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      "uid": "b28ad9afbe483a4746b6179c528aac710706bcf247c47b2c625ec54343e7de1d"
    },
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        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
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        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
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        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
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        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3"
      ],
      "finding_type": "case_review",
      "kafka_offset": null,
      "summary": "15 cited incident observations available. Actor attribution remains separately
unestablished. Missing source timestamps and unverified exploit topology remain limitations. Supported
observations: Deterministic unusual_parent_process signal:
C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe; powershell.exe -NoProfile -ExecutionPolicy
Bypass -EncodedCommand UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=; user=administrator; Command attempts to
clear the Windows Security log; successful deletion is not established.; Executable launched from
ProgramData: C:\\ProgramData\\svchost-helper.exe. Location alone does not establish malware.; PowerShell
executed an encoded command; content and intent require review.; File creation timestamps disagree: SI=2025-
01-01T00:00:00Z, FN=2026-10-13T03:10:00Z. Possible timestomping; copying/restoration must be ruled out.; GCP
service-account key creation observed for svc-backup@specula-lab-synthetic.iam.gserviceaccount.com;
authorization and intent require review.; Source reports access to lsass.exe; completed credential
extraction is not established.; 2 failed logons followed by a successful logon for administrator from
203.0.113.45. Possible credential compromise; brute force is not established by this count alone.",
      "timestamp": "2026-10-13T06:26:54.768354+00:00"
    },
    {
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      "dfkg_refs": [
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        "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
        "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c",
        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
        "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
        "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
        "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130",
        "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9",
        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
        "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3"
      ],
      "finding_type": "case_review",
      "kafka_offset": null,
      "summary": "15 cited incident observations available. Actor attribution remains separately
unestablished. Missing source timestamps and unverified exploit topology remain limitations. Review
legitimate administration, file copying/restoration and source-reported findings as alternative
explanations; temporal order does not establish causation.",
      "timestamp": "2026-10-13T06:26:57.623697+00:00"
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    {
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      "dfkg_refs": [
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        "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d",
        "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b",
        "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
        "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99",
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        "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83",
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        "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2",
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        "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d",
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      ],
      "finding_type": "case_review",
      "kafka_offset": null,
      "summary": "VERDICT: ACCEPT\nConfidence: 0.0\nAccept cited observations for analyst review; no
calibrated probability asserted. 15 cited incident observations available. Actor attribution remains
separately unestablished. Missing source timestamps and unverified exploit topology remain limitations.",
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      "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077",
      "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
      "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
      "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
      "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e",
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        ],
        "summary": "C:\\ProgramData",
        "time": "2025-01-01T00:00:00Z",
        "time_basis": "file_metadata",
        "uid": "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99"
      },
      {
        "attacks": [],
        "canonical_host_id": "host-WS-17",
        "dfkg_refs": [
          "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144"
        ],
        "summary": "user=alice",
        "time": "2026-10-13T03:00:00Z",
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    "time_basis": "recorded_event_time",
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  {
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    "canonical_host_id": "host-WS-17",
    "dfkg_refs": [
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    ],
    "summary": "Source-reported UEBA activity: login; user=alice",
    "time": "2026-10-13T03:00:00Z",
    "time_basis": "recorded_event_time",
    "uid": "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077"
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  {
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    "canonical_host_id": "host-WS-17",
    "dfkg_refs": [
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    ],
    "summary": "user=administrator",
    "time": "2026-10-13T03:05:00Z",
    "time_basis": "recorded_event_time",
    "uid": "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c"
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  {
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    "dfkg_refs": [
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    ],
    "summary": "user=administrator",
    "time": "2026-10-13T03:06:00Z",
    "time_basis": "recorded_event_time",
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    ],
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  },
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    "canonical_host_id": "host-WS-17",
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    ],
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        "summary": "Source-reported UEBA activity: remote_login; user=administrator",
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        "time_basis": "recorded_event_time",
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      },
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        "attacks": [
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        ],
        "canonical_host_id": "host-WS-17",
        "dfkg_refs": [
          "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
        ],
        "summary": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe; powershell.exe -
NoProfile -ExecutionPolicy Bypass -EncodedCommand UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=;
user=administrator",
        "time": "2026-10-13T03:08:00Z",
        "time_basis": "recorded_event_time",
        "uid": "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
      },
      {
        "attacks": [],
        "canonical_host_id": "host-WS-17",
        "dfkg_refs": [
          "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
        ],
        "summary": "C:\\ProgramData\\svchost-helper.exe",
        "time": "2026-10-13T03:10:00Z",
        "time_basis": "recorded_event_time",
        "uid": "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
      },
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        "attacks": [],
        "canonical_host_id": "host-WS-17",
        "dfkg_refs": [
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        ],
        "summary": "user=administrator",
        "time": "2026-10-13T03:11:00Z",
        "time_basis": "recorded_event_time",
        "uid": "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50"
      },
      {
        "attacks": [],
        "canonical_host_id": "host-WS-17",
        "dfkg_refs": [
          "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7"
        ],
        "summary": "svchost-helper.exe; C:\\ProgramData\\svchost-helper.exe; user=administrator",
        "time": "2026-10-13T03:12:00Z",
        "time_basis": "recorded_event_time",
        "uid": "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7"
      },
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        "attacks": [],
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        "dfkg_refs": [
          "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d"
        ],
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```
    "summary": "Unsigned executable launched from ProgramData; user=administrator",
    "time": "2026-10-13T03:12:00Z",
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    "uid": "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d"
  },
  {
    "attacks": [],
    "canonical_host_id": "host-WS-17",
    "dfkg_refs": [
      "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130"
    ],
    "summary": "C:\\ProgramData\\svchost-helper.exe; C:\\ProgramData\\svchost-helper.exe --service",
    "time": "2026-10-13T03:12:00Z",
    "time_basis": "recorded_event_time",
    "uid": "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130"
  },
  {
    "attacks": [
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    "dfkg_refs": [
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    ],
    "summary": "Source-reported memory finding: windows.lsadump",
    "time": "2026-10-13T03:13:00Z",
    "time_basis": "recorded_event_time",
    "uid": "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b"
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  {
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    "time": "2026-10-13T03:13:00Z",
    "time_basis": "recorded_event_time",
    "uid": "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c"
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    "dfkg_refs": [
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    "time": "2026-10-13T03:13:00Z",
    "time_basis": "recorded_event_time",
    "uid": "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83"
  },
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    "attacks": [],
    "canonical_host_id": "host-WS-17",
    "dfkg_refs": [
      "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
    ],
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        "summary": "Source-reported UEBA activity: large_outbound_transfer; user=administrator",
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        "uid": "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
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        "canonical_host_id": "host-WS-17",
        "dfkg_refs": [
          "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
        ],
        "summary": "C:\\Windows\\System32\\cmd.exe; cmd.exe /c wevtutil cl Security; user=SYSTEM",
        "time": "2026-10-13T03:15:00Z",
        "time_basis": "recorded_event_time",
        "uid": "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
      },
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        "attacks": [],
        "canonical_host_id": "gcp-project-specula-lab-synthetic",
        "dfkg_refs": [
          "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c"
        ],
        "summary": "google.iam.admin.v1.CreateServiceAccountKey",
        "time": "2026-10-13T03:20:00Z",
        "time_basis": "recorded_event_time",
        "uid": "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c"
      },
      {
        "attacks": [],
        "canonical_host_id": "gcp-project-specula-lab-synthetic",
        "dfkg_refs": [
          "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c"
        ],
        "summary": "storage.objects.list",
        "time": "2026-10-13T03:22:00Z",
        "time_basis": "recorded_event_time",
        "uid": "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c"
      }
    ],
    "frozen": true,
    "summary": "20 cited evidence events ordered by source time. File metadata dates are labelled and are
not proof of execution time."
  }
}
```


10. Normalized evidence appendix


```
[
  {
    "activity_id": 1,
    "canonical_host_id": "host-LOCAL_HOST",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1001,
```

```
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "file_name": "svchost-helper.exe",
    "file_path": "C:\\ProgramData",
    "fn_created": "2026-10-13T03:10:00Z",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2025-01-01T00:00:00+00:00",
    "security_scan_degraded": true,
    "severity_id": 1,
    "si_created": "2025-01-01T00:00:00Z",
    "source_type": "mft",
    "source_volume_creation_time": "2026-08-01T00:00:00+00:00",
    "time": "2025-01-01T00:00:00Z",
    "trace_id": "ce97e6eb-ffad-4bcc-ad9b-f5ada5e0509c",
    "uid": "4c5ba809fcde8088e5a9f2b7e6dd4edcfdeb541e0b272f8c18ac0d752a737a99"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "logon_type": 2,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:00:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4624,
    "source_outcome": "success",
    "source_type": "evtx",
    "source_user_name": "alice",
    "src_endpoint": {
      "canonical_host_uid": null,
      "ip_address": "10.10.20.44",
      "port": null
    },
    "time": "2026-10-13T03:00:00Z",
    "trace_id": "f2eef222-b6b4-4c0b-bfba-16c643f9c920",
    "uid": "054581ba9d75199bbb67e60c3d8c29a43556fe6ef0dc25077affc76e4daaf144",
    "user": {
      "domain": null,
      "name": "alice",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "analytic_name": "UEBA Engine",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
```

```
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "LastRecordChange": "2026-10-13T03:00:00Z",
      "TimeCreated": "2026-10-13T03:00:00Z",
      "anomaly_name": "Source-reported UEBA activity: login",
      "artifact_type": "ueba_observation",
      "baseline_status": "normal",
      "confidence": "Unspecified",
      "event": "login",
      "eventTime": "2026-10-13T03:00:00Z",
      "host": "WS-17",
      "host_name": "WS-17",
      "risk_score": 5,
      "severity_id": 1,
      "sha256_hash": "f1f6b29cd36ce1d4665084d67da808a7009976b6b18408ef29facf96d4004849",
      "timestamp": "2026-10-13T03:00:00Z",
      "trace_id": "699587d0-d557-4019-9662-a2e1e162d1e6",
      "user": "alice",
      "user_name": "alice",
      "vct_merkle_root": "f93096e7b7d7d827a72a4cc30e113131a696f73189f81abb223e8a03ac6e9c3b",
      "verdict": "normal"
    },
    "finding_title": "Source-reported UEBA activity: login",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:00:00Z",
    "raw_verdict": "normal",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_baseline_status": "normal",
    "source_event_type": "login",
    "source_type": "ueba",
    "source_user_name": "alice",
    "time": "2026-10-13T03:00:00Z",
    "trace_id": "699587d0-d557-4019-9662-a2e1e162d1e6",
    "uid": "f910e776da2f15a86af62a654b0e62ba910fb10952bea236a8100d3bb1588077"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:05:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
```

```
    "source_event_id": 4625,
    "source_outcome": "failure",
    "source_type": "evtx",
    "source_user_name": "administrator",
    "src_endpoint": {
      "canonical_host_uid": null,
      "ip_address": "203.0.113.45",
      "port": null
    },
    "time": "2026-10-13T03:05:00Z",
    "trace_id": "473f6c04-e4a0-43d8-899b-b746e794ed47",
    "uid": "8f2f760fb02d29c238e8bfda6c95f3c6890bd15f1da25aa436839262387d0a9c",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:06:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4625,
    "source_outcome": "failure",
    "source_type": "evtx",
    "source_user_name": "administrator",
    "src_endpoint": {
      "canonical_host_uid": null,
      "ip_address": "203.0.113.45",
      "port": null
    },
    "time": "2026-10-13T03:06:00Z",
    "trace_id": "c498ce7a-f9a3-4647-bee4-8cfa64d1352f",
    "uid": "efcd58b4993ac6ba73c9be4b1c690fa0bdb936517e69763bd494a1e6955ff5a3",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
```

```
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "logon_type": 10,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:07:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4624,
    "source_outcome": "success",
    "source_type": "evtx",
    "source_user_name": "administrator",
    "src_endpoint": {
      "canonical_host_uid": null,
      "ip_address": "203.0.113.45",
      "port": null
    },
    "time": "2026-10-13T03:07:00Z",
    "trace_id": "9874d634-cb5f-475b-8789-8ac52a210917",
    "uid": "05bba0a46369aaa4c1eeb88e26f69813a56707d11de077fb9c80645556dfb5a3",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "analytic_name": "UEBA Engine",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "LastRecordChange": "2026-10-13T03:07:00Z",
      "TimeCreated": "2026-10-13T03:07:00Z",
      "anomaly_name": "Source-reported UEBA activity: remote_login",
      "artifact_type": "ueba_observation",
      "baseline_status": "anomalous",
      "confidence": "Unspecified",
      "event": "remote_login",
      "eventTime": "2026-10-13T03:07:00Z",
      "host": "WS-17",
      "host_name": "WS-17",
      "reasons": [
        "never-before-seen external source",
        "login after repeated failures"
      ],
      "risk_score": 92,
      "severity_id": 1,
```

```
      "sha256_hash": "f9d06156b68c71e871eb4a168bffec3e2c428bbbe0c9eff19afecf8d4b25909d",
      "source_ip": "203.0.113.45",
      "src_ip": "203.0.113.45",
      "timestamp": "2026-10-13T03:07:00Z",
      "trace_id": "cc35d7f2-42e8-43b4-a5a3-d5eb9e9bec65",
      "user": "administrator",
      "user_name": "administrator",
      "vct_merkle_root": "4c2b6d153342866f4c1ef1c2fcd17cae3a726513ef9f4ea6883f6522d8deca6a",
      "verdict": "anomalous"
    },
    "finding_title": "Source-reported UEBA activity: remote_login",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:07:00Z",
    "raw_verdict": "anomalous",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_baseline_status": "anomalous",
    "source_event_type": "remote_login",
    "source_type": "ueba",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:07:00Z",
    "trace_id": "cc35d7f2-42e8-43b4-a5a3-d5eb9e9bec65",
    "uid": "781b78decbed9e6802d1a1a7c9ff22a35d1fa11e8ae74fe201b5372387e5db7e"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1007,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "command_line": "powershell.exe -NoProfile -ExecutionPolicy Bypass -EncodedCommand
UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=",
    "host_name": "WS-17",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "parent_process_name": "C:\\Windows\\System32\\svchost.exe",
    "parent_process_pid": 0,
    "process_name": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
    "process_pid": 0,
    "raw_source_timestamp": "2026-10-13T03:08:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_event_id": 1,
    "source_type": "evtx",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:08:00Z",
    "trace_id": "b4420213-ae89-4385-a687-f2e41aeb93c4",
    "uid": "b36cb5e399cd01384f55048b0973c73722338a8dda75db6579230421053032f9"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1001,
    "clock_skew_offset_ms": 0,
```

```
    "clock_skew_unverified": false,
    "file_name": "svchost-helper.exe",
    "file_path": "C:\\ProgramData\\svchost-helper.exe",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:10:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 11,
    "source_type": "evtx",
    "time": "2026-10-13T03:10:00Z",
    "trace_id": "05d8a32c-4f86-448c-8295-d3226cd335fe",
    "uid": "fdb738510e16cf048f7e2e05ba0178ee3656dffc97942168fe072fe30277ee1d"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 3,
    "class_uid": 3002,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "dst_endpoint": {
      "canonical_host_uid": "host-WS-17",
      "ip_address": null,
      "port": null
    },
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:11:00Z",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_event_id": 4672,
    "source_type": "evtx",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:11:00Z",
    "trace_id": "23188401-e454-4e98-b12d-5ae9f290ab52",
    "uid": "5cc98f8e3e7080e98a6da4b6c645398ef60b136afdd9034094c5ad99f17bfd50",
    "user": {
      "domain": null,
      "name": "administrator",
      "sid": null
    }
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-e7cc792584f84b4792553c73cd53c73d",
    "category_uid": 1,
    "class_uid": 1007,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "file_path": "C:\\ProgramData\\svchost-helper.exe",
    "host_name": "WS-17",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "parent_process_pid": 0,
    "process_name": "svchost-helper.exe",
    "process_pid": 0,
```

```
    "raw_source_timestamp": "2026-10-13T03:12:00Z",
    "security_scan_degraded": false,
    "severity_id": 4,
    "source_event_type": "process_start",
    "source_finding": "Unsigned executable launched from ProgramData",
    "source_type": "edr",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:12:00Z",
    "trace_id": "cd659720-9d61-4f9f-a9f8-d5c49fb0ce2e",
    "uid": "05684568d548b24e017081896647594bd348fb547838aee72f47b5be10a01dc7",
    "user_name": "administrator"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "High",
    "finding_title": "Unsigned executable launched from ProgramData",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:12:00Z",
    "security_scan_degraded": false,
    "severity_id": 4,
    "source_event_type": "process_start",
    "source_finding": "Unsigned executable launched from ProgramData",
    "source_type": "edr",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:12:00Z",
    "trace_id": "60288c97-9df8-4e49-a331-b619035f7ec0",
    "uid": "2036589e8359f589777d00d10fd3de12c9f753ffadb6ce9c5dd9b33723981f4d"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1007,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "command_line": "C:\\ProgramData\\svchost-helper.exe --service",
    "host_name": "WS-17",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "parent_process_name": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
    "parent_process_pid": 0,
    "process_name": "C:\\ProgramData\\svchost-helper.exe",
    "process_pid": 0,
    "raw_source_timestamp": "2026-10-13T03:12:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_event_id": 1,
    "source_type": "evtx",
    "time": "2026-10-13T03:12:00Z",
    "trace_id": "06dbb6d6-2d35-44f5-9a3c-db3f6e16c599",
    "uid": "ae45aad54191c74830219f20d0ea25e8b17b1cbab46cb18953e984d7f23f0130"
```

```
  },
  {
    "activity_id": 1,
    "analytic_name": "Volatility windows.lsadump",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "LastRecordChange": "2026-10-13T03:13:00Z",
      "TimeCreated": "2026-10-13T03:13:00Z",
      "eventTime": "2026-10-13T03:13:00Z",
      "finding": "Credential material pattern observed in memory snapshot",
      "host": "WS-17",
      "host_name": "WS-17",
      "note": "Synthetic finding only; no real credentials present",
      "plugin": "windows.lsadump",
      "severity": "critical",
      "severity_id": 5,
      "sha256_hash": "996da47417d742ba5b8aa65c177d0f02ce0d496f3aedea3beca1b441756f4ff3",
      "timestamp": "2026-10-13T03:13:00Z",
      "trace_id": "06b3adcb-4e6a-4448-8adb-a2f82284fff3",
      "vct_merkle_root": "5d21ffa6395177977223d673b13865dabd01282ea85572e34728a9160459a0c7"
    },
    "finding_title": "Source-reported memory finding: windows.lsadump",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:13:00Z",
    "raw_verdict": "Credential material pattern observed in memory snapshot",
    "security_scan_degraded": false,
    "severity_id": 5,
    "source_finding": "Credential material pattern observed in memory snapshot",
    "source_type": "memory",
    "time": "2026-10-13T03:13:00Z",
    "trace_id": "06b3adcb-4e6a-4448-8adb-a2f82284fff3",
    "uid": "351132c3f184207bb57a1f7ba128210d29db2979dd96057b1d546aff17a8e34b"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "access_mask": "0x1010",
      "attack_indicator": "credential dumping behavior",
      "event_type": "credential_access",
      "host": "WS-17",
      "operation": "OpenProcess",
      "process": "svchost-helper.exe",
      "severity": "critical",
      "sha256_hash": "3657b111b92214eaf6b42430d11fa883c3b410d792aa7a634c37fc39103d103a",
      "target_process": "lsass.exe",
```

```
      "timestamp": "2026-10-13T03:13:00Z",
      "trace_id": "aeff10aa-0b18-4fc3-9dd4-05b0b9b30103",
      "vct_merkle_root": "e3fc26a4b936c28eb5ab765e8d469dad3e97db5f1226e37d4e7cb8d81a089b7e"
    },
    "finding_title": "Source-reported credential access",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:13:00Z",
    "security_scan_degraded": false,
    "severity_id": 5,
    "source_event_type": "credential_access",
    "source_type": "edr",
    "time": "2026-10-13T03:13:00Z",
    "trace_id": "aeff10aa-0b18-4fc3-9dd4-05b0b9b30103",
    "uid": "633a1287fa75a429bf974b458dd34f07ffa077d51a1d58738e31dea2142db58c"
  },
  {
    "activity_id": 1,
    "analytic_name": "Volatility windows.malfind",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "ImageFileName": "svchost-helper.exe",
      "LastRecordChange": "2026-10-13T03:13:00Z",
      "PID": 4312,
      "TimeCreated": "2026-10-13T03:13:00Z",
      "eventTime": "2026-10-13T03:13:00Z",
      "finding": "RWX private memory region with PE-like header",
      "host": "WS-17",
      "host_name": "WS-17",
      "pid": 4312,
      "plugin": "windows.malfind",
      "process": "svchost-helper.exe",
      "process_name": "svchost-helper.exe",
      "process_pid": 4312,
      "severity": "high",
      "severity_id": 4,
      "sha256_hash": "247f3c3322f5d3f164ac52f50773f841077ef579d10dccde7314d747ff928b96",
      "timestamp": "2026-10-13T03:13:00Z",
      "trace_id": "bd1687f5-3a34-4811-9897-762450cfa2b5",
      "vct_merkle_root": "d7cb0fafef5c4ae2c42fca3ca2c176ad4a4eba969f416615405d71dfc76dfc3e"
    },
    "finding_title": "Source-reported memory finding: windows.malfind",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:13:00Z",
    "raw_verdict": "RWX private memory region with PE-like header",
    "security_scan_degraded": false,
    "severity_id": 4,
    "source_finding": "RWX private memory region with PE-like header",
    "source_type": "memory",
    "time": "2026-10-13T03:13:00Z",
    "trace_id": "bd1687f5-3a34-4811-9897-762450cfa2b5",
    "uid": "7285659d689ffcae6f68e166bc4d3d52275818f7cdb9ad10975b806e31fd1c83"
```

```
  },
  {
    "activity_id": 1,
    "analytic_name": "UEBA Engine",
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-c653ec6838174caabe0845b9a723dda8",
    "category_uid": 2,
    "class_uid": 2004,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "confidence": "Unspecified",
    "details": {
      "LastRecordChange": "2026-10-13T03:14:00Z",
      "TimeCreated": "2026-10-13T03:14:00Z",
      "anomaly_name": "Source-reported UEBA activity: large_outbound_transfer",
      "artifact_type": "ueba_observation",
      "baseline_status": "anomalous",
      "bytes": 2457600,
      "confidence": "Unspecified",
      "destination_ip": "198.51.100.77",
      "dst_ip": "198.51.100.77",
      "event": "large_outbound_transfer",
      "eventTime": "2026-10-13T03:14:00Z",
      "host": "WS-17",
      "host_name": "WS-17",
      "risk_score": 88,
      "severity_id": 1,
      "sha256_hash": "36c535ee3c88475f6d0a563266c91f4da94f95c69409b9f00ca67681d4fe3f93",
      "timestamp": "2026-10-13T03:14:00Z",
      "trace_id": "6ffd9f32-3ac0-47ff-bcdb-8eda321a6bce",
      "user": "administrator",
      "user_name": "administrator",
      "vct_merkle_root": "606e6ad44c98b9e3da78ecaab467c9b7d7b4e21427124f403b6c32357cfe431e",
      "verdict": "anomalous"
    },
    "finding_title": "Source-reported UEBA activity: large_outbound_transfer",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:14:00Z",
    "raw_verdict": "anomalous",
    "security_scan_degraded": false,
    "severity_id": 1,
    "source_baseline_status": "anomalous",
    "source_event_type": "large_outbound_transfer",
    "source_type": "ueba",
    "source_user_name": "administrator",
    "time": "2026-10-13T03:14:00Z",
    "trace_id": "6ffd9f32-3ac0-47ff-bcdb-8eda321a6bce",
    "uid": "abeca332837ad018033a2b39eb19e6ead03fa85a86e668c800f61f0b4f1e42b2"
  },
  {
    "activity_id": 1,
    "canonical_host_id": "host-WS-17",
    "case_id": "LOCAL-67f1747f4f914268b409082cafac3155",
    "category_uid": 1,
    "class_uid": 1007,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": false,
    "command_line": "cmd.exe /c wevtutil cl Security",
```

```
    "host_name": "WS-17",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "parent_process_name": "C:\\ProgramData\\svchost-helper.exe",
    "parent_process_pid": 0,
    "process_name": "C:\\Windows\\System32\\cmd.exe",
    "process_pid": 0,
    "raw_source_timestamp": "2026-10-13T03:15:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_event_id": 1,
    "source_type": "evtx",
    "source_user_name": "SYSTEM",
    "time": "2026-10-13T03:15:00Z",
    "trace_id": "7673503c-e3b6-4049-a769-24f85e55882c",
    "uid": "e89078174b2b87ff91c46c988277bbe798efb3db976fe4b6e69c2d07c528c02d"
  },
  {
    "activity_id": 1,
    "api_operation": "google.iam.admin.v1.CreateServiceAccountKey",
    "canonical_host_id": "gcp-project-specula-lab-synthetic",
    "case_id": "LOCAL-e7cc792584f84b4792553c73cd53c73d",
    "category_uid": 6,
    "class_uid": 6003,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "cloud_account_id": "specula-lab-synthetic",
    "cloud_provider": "GCP",
    "cloud_region": "global",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:20:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_ip": "203.0.113.45",
    "source_type": "gcp_audit",
    "time": "2026-10-13T03:20:00Z",
    "trace_id": "c11a8b82-0ee1-47b4-a02f-6e4822d5f0d8",
    "uid": "4af97cc18a83f61584a65218feb0d85cd210b7ce93560fc46d97ba7fa068445c",
    "user_identity": "svc-backup@specula-lab-synthetic.iam.gserviceaccount.com"
  },
  {
    "activity_id": 1,
    "api_operation": "storage.objects.list",
    "canonical_host_id": "gcp-project-specula-lab-synthetic",
    "case_id": "LOCAL-e7cc792584f84b4792553c73cd53c73d",
    "category_uid": 6,
    "class_uid": 6003,
    "clock_skew_offset_ms": 0,
    "clock_skew_unverified": true,
    "cloud_account_id": "specula-lab-synthetic",
    "cloud_provider": "GCP",
    "cloud_region": "global",
    "is_summary": false,
    "ocsf_version": "1.2.0",
    "raw_source_timestamp": "2026-10-13T03:22:00Z",
    "security_scan_degraded": true,
    "severity_id": 1,
    "source_ip": "203.0.113.45",
```

```
    "source_type": "gcp_audit",
    "time": "2026-10-13T03:22:00Z",
    "trace_id": "ec0d789c-3829-4b74-8867-63621ddda158",
    "uid": "060d9078794c40e68bd9445d55b09da7185c00f06748f44a311177a2f206853c",
    "user_identity": "svc-backup@specula-lab-synthetic.iam.gserviceaccount.com"
  }
]
```


11. Cyber-team follow-up

1. Review cited raw artifacts and confirm source clock/custody assumptions.
2. Validate detections against additional endpoint/network telemetry and document counter-evidence.
3. Confirm indicators and scope before containment or remediation.
4. Re-run with missing telemetry and retain the input hashes, selections and software fingerprints.

12. Limitations and alternative explanations


```
[
  "actor_sequence_inferred",
  "actor_sequence_unavailable",
  "security_screening_degraded",
  "semantic_mapping_unavailable",
  "source_clock_skew_unverified",
  "unsupported_technique_id",
  "uploaded_record_coverage_limited"
]
```


Unsupported records, source selection, excluded ports and missing telemetry can reduce coverage. Model inference, attribution similarity and event ordering do not establish actor identity, intent or causality. Collection integrity and error rates require independent validation.

13. Examiner and independent reviewer approval

Examiner name / qualifications: _____

Scope and methods reviewed: _____

Examiner signature / date: _____

Independent reviewer signature / date: _____

Approval status: UNSIGNED. No signature or certification is generated by this system.

Specula examination template v1 • Case LOCAL-c653ec6838174caabe0845b9a723dda8 • End of report